
A Training-Time Sign Flip in IOI Circuit Formation

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Abstract

A model can depend on the same internal state in opposite ways at different stages of training. Circuit analyses usually describe a trained model at a single checkpoint, but the role a component plays can differ earlier, and we show a case where a state that is harmful to a behavior early becomes useful for it later. We study indirect object identification, the task of completing a sentence such as “When Mary and John went to the store, John gave a drink to” with the name that occurred once (Mary) rather than the repeated name (John). During training, accuracy on this two-way choice falls below chance in a transient window, while the language modeling loss on a fixed Pile training-stream sample keeps decreasing. Replacing the internal representation at the repeated name’s second occurrence with a matched non-repeating one improves the choice during the window and degrades it after recovery, so the model’s dependence on that state reverses across training. This positive-to-negative reversal replicates across six model scales from 160 million to 12 billion parameters, across nine training runs that vary seed, data order, and initialization, and in a second model family. During recovery a head that attends to the repeated name and lowers its output score develops a large causal effect, after the harmful dependence on the S2 state is already present. Separately, the repeated name signal becomes linearly decodable on its own timeline, and removing the direction a probe uses to read it does not reduce the model’s preference for the repeated name.

1. Introduction

During training a language model can pass through a window in which it performs below chance on a structured task,

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actively preferring the wrong answer, and then recover to near-perfect accuracy. The window is not apparent at the five sampled points of an evaluation-loss trajectory, which decreases throughout, so a study of the loss curve alone would miss it. This raises a question about how we read circuits. Mechanistic analyses typically describe a model at one mature checkpoint, but the circuits that implement a behavior are not present at initialization. They form over training (Elhage et al., 2021; Olsson et al., 2022), and a state that supports the behavior at maturity need not have supported it earlier.

We study this in indirect object identification (IOI), a controlled name tracking task. Given a sentence such as “When Mary and John went to the store, John gave a drink to”, a competent model continues with the indirect object Mary, the name that appears once, rather than the repeated subject John, the name that appears twice. We write the second occurrence of the repeated subject as S2, and we score the model by the logit difference between the two candidate names, defined as the logit of the correct name minus the logit of the repeated name. A positive logit difference means the model prefers the correct name and a negative logit difference means it prefers the repeated name. Below chance refers to this two-way choice between the two names, not to the model’s ranking over the full vocabulary.

Reading this behavior across training means keeping three questions apart, because it is easy to treat them as the same question. The first is whether the information the task needs is present in the model’s internal state, which we test by training a simple classifier to read that state. The second is whether the state at a particular position actually affects the output, which we test by replacing that state and watching how the output changes. The third is whether the model has built a working part that carries out the task, which we test by removing a candidate part and measuring how much the behavior degrades. The accessibility of a signal, the causal role of a position, and the formation of a component can be conflated. The central result of this paper is that during the formation of the indirect object identification behavior they appear at different times.

We report three findings. First, IOI accuracy falls below chance in a transient training window and recovers, and this appears across model scales, random seeds, data orders,

initializations, and two model families. Second, a matched intervention at S2 improves the two-way choice during the window and harms it after recovery, so the model’s dependence on the S2 state reverses across training, and this reversal replicates across all six Pythia scales, all nine PolyPythias training variants, and Stanford GPT-2 Small. Third, a suppressor head that attends to the repeated name and lowers its logit acquires a large causal effect during recovery, after the harmful dependence on the S2 state is already present, while the direction a probe uses to read the repeated name is not sufficient to explain the model’s preference for it.

2. Related Work

The IOI circuit in GPT-2 Small was reverse engineered by Wang et al. (2023), who named its functional parts, including name mover heads that copy the correct name and suppressor heads that attend to the repeated name and write against it. Later work automated circuit discovery (Conmy et al., 2023) and set out the framework in which attention heads compose (Elhage et al., 2021). Simpler mechanisms such as induction heads form at identifiable points in training (Olsson et al., 2022), and internal representations can change before behavior does (Nanda et al., 2023). Tigges et al. (2024) track the IOI circuit across Pythia training and scale, report that task performance is generally non-decreasing once a task is acquired, and note that task scores can become negative immediately before a task is learned, leaving the cause open. We isolate this pre-recovery regime in IOI, show that the repeated subject outranks the correct name within the full vocabulary during it, and identify a position-level intervention whose effect changes sign across recovery.

Our phenomenon is distinct from two effects it might be confused with. A model can generalize long after its training loss saturates, a pattern studied as grokking (Power et al., 2022; Nanda et al., 2023). Separately, apparent jumps in capability with scale have been attributed to a discontinuous metric rather than to a change in the model (Wei et al., 2022; Schaeffer et al., 2023). The window we describe differs from both, because the measured loss decreases at every sampled checkpoint through it and the reversal appears in the continuous logit difference, a signed quantity, and not only in a thresholded score. Our measurements use three standard tools applied across training, linear probes (Alain & Bengio, 2017; Belinkov, 2022), activation replacement following causal mediation analysis and activation patching (Vig et al., 2020; Meng et al., 2022; Goldowsky-Dill et al., 2023; Zhang & Nanda, 2024), and head ablation (Wang et al., 2023).

3. Experimental Setup

Task. For the primary behavior and residual-patching analyses, we use the IOI templates of Wang et al. (2023), restricted to ten single-token name templates in balanced ABBA and BABA order, with 30 prompts per template for 300 prompts in total. An ABBA template introduces the indirect object first and a BABA template introduces the repeated subject first, and we keep the two orders balanced so that position alone does not decide the answer. Single-token names keep the two-way choice unambiguous, since one next-token prediction decides each prompt. We report the logit difference defined above and the fraction of prompts with a positive logit difference as accuracy.

Models. We analyze released Pythia checkpoints (Biderman et al., 2023) at six scales from 160 million to 12 billion parameters, nine PolyPythias 160M variants that independently vary seed, data order, and initialization (van der Wal et al., 2025), and Stanford GPT-2 Small (Karamcheti et al., 2021). Our detailed intervention and suppressor analyses use Pythia 160M. We also train a Pythia 160M model from scratch with a new seed, which reproduces the below-chance window and provides an additional check that the window is not unique to the released checkpoint series.

Matched S2 intervention. For each prompt we build a matched control that is identical except that the second occurrence of the repeated name is replaced by a third, unrelated single-token name. We then replace the internal representation at S2 over a middle layer window in the original prompt with the representation from the control, leave every other position unchanged, and measure the change in logit difference. This follows causal mediation analysis and activation patching (Vig et al., 2020; Meng et al., 2022; Goldowsky-Dill et al., 2023), with the metric conventions discussed by Zhang & Nanda (2024). The window is layers 3 to 5 for Pythia 160M and is scaled to each model’s depth for the other scales; the six windows are listed in Appendix E. A positive change means the intervention moves the model toward the correct name. The procedure is fixed across training, and the state it inserts is drawn from each checkpoint’s own control, so it changes from checkpoint to checkpoint.

Head ablation. We use two head-level interventions and keep them distinct. For the localization sweep during the window, we zero-ablate each attention head, meaning we set its output to zero at a checkpoint, and we measure the change in logit difference. For the suppressor trajectory, we mean-ablate the selected head, meaning we replace its output with its average output over the evaluation set, and we measure the change in logit difference. In both cases the reported quantity is a counterfactual behavioral effect, not an additive contribution.

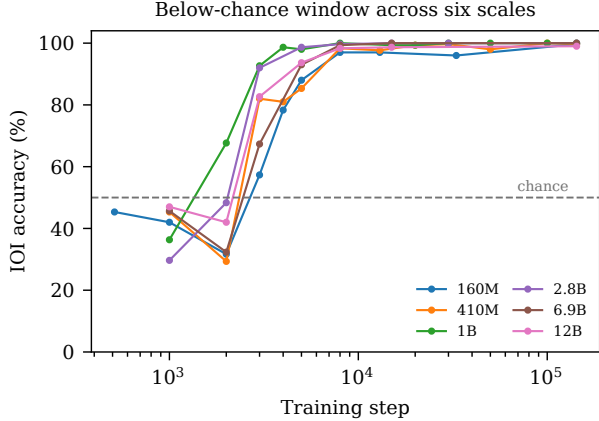


Figure 1. IOI accuracy across training for six Pythia scales, measured under the ten-template protocol used for the cross-scale behavior and residual-patching analyses. Accuracy falls below the two-way chance level of 50 percent and recovers at every scale. Floor accuracies range from 29.3 percent at 410 million parameters to 42.0 percent at 12 billion parameters.

Probes and statistics. We train logistic regression probes (Alain & Bengio, 2017; Belinkov, 2022) to predict, from the representation at S2, whether a name is repeated, and we evaluate them on names and templates that were not used for training the probe. For intervention analyses, uncertainty is reported as 95 percent bootstrap intervals over prompts unless otherwise stated. For the primary Pythia 160M intervention we also resample by template and by name-pair, and the effect stays on the same side of zero under both, as reported in Appendix E.

4. A Below-Chance Preference Reversal

IOI accuracy is non-monotonic during training. After an initial rise it falls below the two-way baseline of 50 percent and then recovers to near-perfect accuracy. On Pythia 160M it reaches 31.7 percent accuracy at step 2000 before recovering. The window appears at every scale we test from 160 million to 12 billion parameters, with floor accuracies from 29.3 to 42.0 percent (Figure 1 and Appendix E), in Stanford GPT-2 Small, and in all nine PolyPythias variants, whose floor accuracies range from 14.7 to 36.7 percent (Appendix F). Because the PolyPythias variants separately vary seed, data order, and initialization, the window is not the product of a single unlucky run.

During this window the model has not simply lost the ability to produce an answer. At the Pythia 160M floor its most likely next word is a token other than either candidate on 98.67 percent of prompts, usually a common function word such as “the”. The reversal shows up only when the two names are compared directly. In that comparison the repeated name has the higher probability, 0.0238 against

0.0118 for the correct name, and the better position in the full vocabulary, an average rank of 14.9 against 29.4. So the model rarely tries to say the repeated name out loud, yet when the two names are placed side by side it scores the wrong one higher.

The same reversal appears in the continuous logit difference as well as in the accuracy that thresholds it. Across the floor interval the logit difference is negative, with bootstrap intervals that exclude zero, so over a contiguous stretch of training the model gives the repeated name a higher score than the correct name. Because this is a movement in a signed quantity, it reflects a genuine change in which name the model prefers and does not come from rounding a close score to a win or a loss. Throughout the same stretch, the language modeling loss on a fixed sample from the Pile training stream (Gao et al., 2020) decreases at every sampled checkpoint, so the window is not apparent in the measured loss trajectory (Appendix A).

5. The Causal Role of S2 Reverses

The matched S2 intervention changes sign across training (Figure 2, center). During the window, replacing the S2 representation with its matched non-repeating control raises the logit difference by 0.95 at step 2000, moving the model toward the correct name. The intervention is strongly positive at step 2000, indistinguishable from zero at step 3000, where its 95 percent interval spans roughly negative 0.11 to positive 0.15, and significantly negative by step 4000, where it is about negative 0.95. It reaches negative 4.12 at the final checkpoint. The model’s dependence on the S2 state therefore reverses over training, from harmful during the window to useful at maturity. This primary result survives dependence-aware inference. Resampling by template places the dip effect in the interval 0.68 to 1.25 and the maturity effect in negative 4.54 to negative 3.72, and resampling by name-pair gives similar intervals, all on the same side of zero (Appendix E).

This reversal is not confined to one model. The positive-to-negative reversal replicates across all six Pythia scales from 160 million to 12 billion parameters, across all nine PolyPythias training variants, and in Stanford GPT-2 Small (Appendices E, F, and G).

Among the positions tested, only S2 perturbations reverse sign. Table 1 compares matched and random perturbations at S2 with random perturbations at the first occurrence, the correct name position, and a structural position. Only the two S2 interventions move from significantly positive during the window to significantly negative at maturity. Replacing the first occurrence of the repeated name raises the logit difference at both stages. Perturbing the correct name or a structural position lowers it at both stages. Some of these

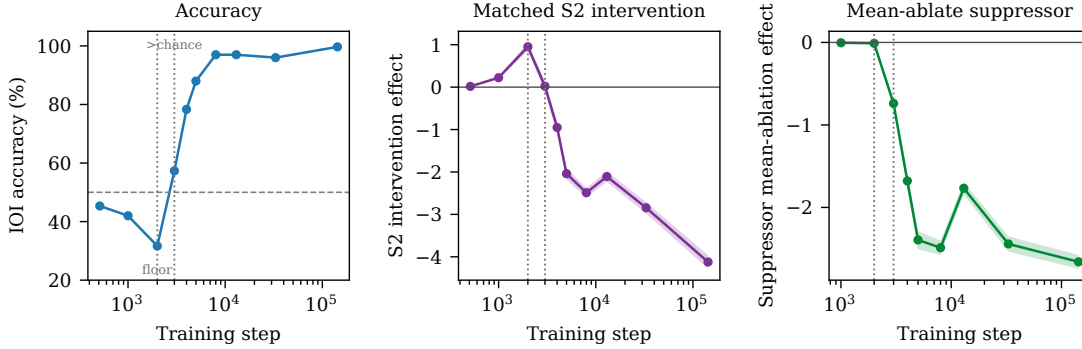


Figure 2. Three measurements on Pythia 160M across training, sharing the training step axis, with the transition window from step 2000 to 3000 shaded. Left, IOI accuracy with the chance line. Center, the effect of the matched S2 intervention on the logit difference, with a 95 percent bootstrap band; a positive value means the intervention moves the model toward the correct name. Right, the effect of mean-ablating the mature suppressor head, where a negative value means removing the head lowers the logit difference, so the head helps the correct name. The center and right panels use opposite sign conventions. At the floor the S2 intervention helps and mean-ablating the suppressor has little effect. After recovery the S2 intervention harms and mean-ablating the suppressor is increasingly damaging. These changes unfold over the broader recovery period but do not by themselves establish that suppressor formation causes the crossover.

Table 1. Effect of perturbing four positions on the logit difference, during the window (step 2000) and at maturity (final checkpoint), on Pythia 160M. A positive value moves the model toward the correct name. Only the two S2 perturbations change sign across training. The full battery with intervals is in Appendix B.

Position and type	Window	Maturity	Reverses
S2 matched name	+0.95	-4.12	yes
S2 random	+0.92	-4.13	yes
S1 random	+2.15	+2.08	no
Correct name random	-3.54	-9.06	no
Structural random	-0.81	-4.45	no

effects are larger in magnitude than the S2 effect, so what distinguishes the S2 position is the change of sign across training rather than the size of the effect at any single checkpoint. A norm-matched random perturbation at S2 reverses sign just as the matched replacement does. The reversal is therefore not specific to the matched control representation, and, as we show in Section 7, the direction a probe uses to read the repeated name is not sufficient to explain it either.

An input-level control supports the same reading from the input side. On a locked Pythia 160M benchmark of 800 prompts with a fixed prompt set, de-duplicating the repeated name raises the logit difference by 0.833, with a cluster-aware interval of 0.681 to 1.023, while a placebo that changes an unrelated filler token has an effect of only negative 0.082, an order of magnitude smaller. An alternate de-duplication with a different replacement name gives 0.832, matching the standard one. This is a different intervention from the residual patching above, so we treat it as complementary evidence (Appendix B).

6. A Mature Suppressor Emerges During Recovery

No individual attention head has a large zero-ablation effect during the window. At step 2000 the largest change in logit difference from zero-ablating any single head, across all 144 heads, is 0.06. The source of the preference for the repeated name during the window is therefore not localized by a head-level analysis.

A suppressor head acquires its effect later. On Pythia 160M we select the head whose mean ablation most reduces mature IOI performance, which is L8H9, and we verify that it attends to the second occurrence of the repeated name and lowers the repeated name’s logit far more than the correct name’s logit. Mean-ablating L8H9 at maturity lowers the logit difference by 2.66, the largest single-head effect in the model, and the dominant head in Stanford GPT-2 Small is likewise a suppressor. Tracing L8H9 across training, mean-ablating it has only a very small effect during the window, a change of negative 0.01 at step 2000, and this becomes increasingly negative during recovery, reaching negative 0.74 at step 3000 and negative 2.39 at step 5000, so removing the head increasingly harms the correct name (Figure 2, right, and Appendix D). The dominant head in Pythia 410M is L12H12, with the same pattern.

Because L8H9 is selected using its mature effect, its early trajectory could in principle reflect the way it was chosen. To guard against this, we repeat the analysis with a mature-defined suppressor set that is frozen before evaluation and tested on 192 held-out items, giving 384 matched contrasts per checkpoint, with cluster-aware intervals. At steps 1800 and 2000 replacing the native activations of this set improves the fair three-way margin, by 0.091 at step 2000 with an

interval from 0.062 to 0.121, which shows that the mature-defined set is not yet functioning suppressively. Its effect is indistinguishable from zero at the intermediate checkpoints and is significantly negative by step 6000, where replacing the set changes the margin by negative 0.64 with an interval from negative 0.80 to negative 0.47. This is a different intervention and a component-level estimand, so we do not describe it as the same sign flip as the S2 residual patch. A separate single-head analysis, using one mature-selected head on a disjoint 800 example set, likewise finds no detectable suppressive effect at step 2000, with an effect of 0.04 and an interval from negative 0.02 to 0.11. The frozen-set trajectory places robust suppressor function after the window, while the disjoint single-head analysis independently confirms that no clearly detectable suppressive effect is present at step 2000 (Appendix D).

The suppressor’s causal effect grows during recovery, after the harmful S2 dependence is already present. Its attention to the second occurrence of the repeated name and its direct contribution against that name both rise during recovery, and its mean-ablation effect grows over the interval in which the S2 intervention moves from positive to negative. These measurements are consistent with the suppressor contributing to recovery. They do not establish that the suppressor’s formation is what causes the crossover. Suppressor function develops during the broader recovery phase, after harmful dependence on the S2 state is already detectable, and we have not traced a single path that carries the early signal from S2 to the suppressor.

7. Accessibility Is Distinct from Causal Function

The repeated name signal becomes linearly decodable during training. The probe reads the S2 representation and predicts whether that position holds a repeated name, as in the original prompt, or a fresh non-repeating name, as in the matched control. Trained on one set of names and templates and evaluated on a disjoint set, it does not generalize above chance in its trained label orientation at initialization, then reaches about 61 percent during the window and about 99 percent at maturity. Below chance accuracy at initialization means the probe does not transfer in the orientation it learned, not that no information exists, and the rise over training is what shows a transferable signal being learned. Per layer values and the split are in Appendix C.

Part of the high accuracy that a probe reaches on names it has already seen comes from the fixed structure of the network rather than from a learned signal. In a separate control we shuffle the token positions before probing. Across middle layers at initialization the intact probe reaches about 88 to 90 percent and shuffling reduces it by roughly 6 to 10 points, so positional structure contributes materially to the high

initialization result (Appendix C).

Decodability does not identify the state that drives the window. In a separate projection experiment, we train a probe on the S2 representation and then remove the direction it uses. Removing this direction does not reduce the model’s preference for the repeated name, and in fact slightly increases it, while removing unrelated control directions has almost no effect (Appendix H). The direction that decodes the repeated name is therefore not sufficient to explain the preference for it. Linear accessibility, the causal role of the S2 state, and the formation of the suppressor follow three different timelines and cannot be treated as one event.

8. Discussion

The central finding is that the causal role of an internal state is not fixed during training. The S2 state that is useful for correct behavior at maturity is one whose perturbation improves behavior during an earlier window. This extends the observation that internal representations can shift before behavior changes (Nanda et al., 2023). In the IOI case the S2 state supports the repeated name during the window and supports the correct name after recovery, so its causal contribution reverses rather than simply coming online.

These results separate three notions that are often merged. Linear accessibility, measured by a probe, position-level causal dependence, measured by replacing a state, and the formation of a task-solving component, measured by removing a head, occur on distinct timelines during IOI formation. A description that is faithful at maturity can therefore be wrong about an earlier checkpoint, which argues for indexing circuit descriptions by training stage.

Our study has clear limits. It does not identify the complete source of the preference for the repeated name during the window, which a head-level analysis does not localize, and we do not claim that the S2 state is a dedicated detector of repetition. The matched S2 intervention is a fixed procedure whose inserted state is drawn from each checkpoint’s own control, so what reverses across training is the model’s dependence on the S2 state, and we do not claim that a single fixed activation changes its role. We study one task, decoder-only models, and English templates. Whether similarly sign-changing causal dependencies occur in other circuits remains open.

Software and Data

Code, source result files, checkpoint identifiers, and analysis scripts for the reported experiments are available in the public repository linked from the camera-ready version. Appendix A documents the one historical evaluation sample that cannot be reconstructed exactly from the stored artifact.

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A. The Below-Chance Window and the Loss

Figure 3 plots the language modeling loss of Pythia 160M on a fixed sample from the Pile training stream (Gao et al., 2020) alongside its IOI accuracy across training. The evaluation uses the monology Pile uncopyrighted dataset on its train split. The producing script skips an initial block of the stream, reads a fixed number of texts, keeps those with at least 512 tokens, truncates each to 512 tokens, and averages the causal language modeling loss over the first 50 qualifying sequences, at steps 1000, 2000, 3000, 5000, and 10000. Because this is the train split, we do not describe it as a held-out set. The loss decreases at every sampled checkpoint over the same interval in which accuracy falls below chance and recovers. Five sampled checkpoints cannot rule out every small feature of the loss curve; they establish that the below-chance window is not apparent in the measured loss trajectory. The script records the procedure and the aggregate losses but not the identities of the sampled texts, so the exact sample is not reconstructable from the stored artifact.

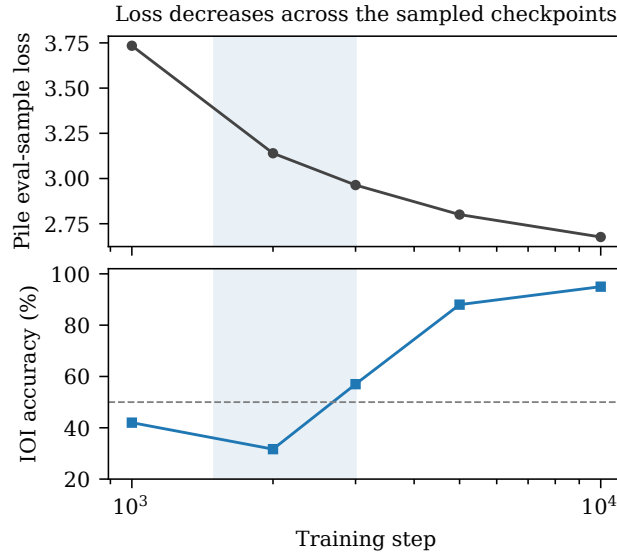


Figure 3. Pile evaluation-sample loss and IOI accuracy for Pythia 160M across training, shown as two aligned panels. The loss decreases at every sampled checkpoint while accuracy dips below chance near step 2000 and recovers.

B. Full Control Battery and Input-Level Control

Table 2 reports every arm of the position battery with 95 percent bootstrap intervals, on Pythia 160M, at the window (step 2000) and at maturity. The two S2 arms are the matched name replacement and a random perturbation of the same vector norm at the same position. Among the tested positions, only the two S2 arms are significantly positive during the window and significantly negative at maturity. The other arms have effects that are often large but keep the same sign at both stages, so a change of sign is specific to the S2 position and does not follow from perturbing the other tested positions. As a narrow sanity check, a perturbation at a non-repeating first position produces a change of 0.000 at the window; this control is weak on its own, because the target and donor states at a non-repeating position can be nearly identical, and the position battery is the stronger evidence against generic disruption.

Table 3 reports a complementary input-level control on a locked Pythia 160M benchmark of 800 prompts with a fixed prompt set, identified by a stored prompt hash, and with cluster-aware intervals. De-duplicating the repeated name raises the logit difference, an alternate de-duplication with a different replacement name gives an almost identical effect, and a placebo that changes an unrelated filler token has an effect an order of magnitude smaller. The clean accuracy under this benchmark is 30.9 percent, with a cluster-aware interval of 23.7 to 38.3 percent. Under the earlier prompt-level protocol the same input control gives de-duplication effects of 0.185 at 410 million parameters and 0.544 at 1 billion parameters, with placebo effects of 0.090 and 0.106, as additional point estimate replications.

Table 2. Full position battery on Pythia 160M. Change in logit difference with 95 percent bootstrap intervals. A positive value moves the model toward the correct name. Only the S2 arms reverse sign between the window and maturity.

Arm	Window (step 2000)		Maturity (final)		Reverses sign
S2 matched name replacement	+0.95	[+0.86, +1.05]	−4.12	[−4.28, −3.96]	yes
S2 random perturbation	+0.92	[+0.84, +1.00]	−4.13	[−4.31, −3.94]	yes
First occurrence random	+2.15	[+2.06, +2.24]	+2.08	[+1.88, +2.27]	no
Correct name random	−3.54	[−3.73, −3.34]	−9.06	[−9.31, −8.82]	no
Structural position random	−0.81	[−0.90, −0.73]	−4.45	[−4.67, −4.23]	no

Table 3. Input-level control at the window on the locked Pythia 160M benchmark, 800 prompts, with cluster-aware 95 percent intervals. The two de-duplications agree and the placebo effect is an order of magnitude smaller.

Intervention	Effect on logit difference	Cluster-aware interval or comparison
Standard de-duplication	+0.833	[+0.681, +1.023]
Alternate de-duplication	+0.832	agrees with standard
Filler placebo	−0.082	[−0.159, −0.012]

C. Probe Experiments

We report three separate probe experiments and keep them distinct.

Held-out generalization. The probe reads the S2 representation and predicts a binary label. The positive class is the S2 position in the original prompt, where the name is a repeat of the earlier subject. The negative class is the same position in a matched control, where that name is replaced by a third, non-repeating single-token name. The probe is trained on 600 examples drawn from one set of names and templates and evaluated on a disjoint set of names and templates, in two folds that swap the training and evaluation sets, and we report the mean over folds per layer without selecting a best layer on validation data. Table 4 reports held-out accuracy at two layers at three training steps. The held-out accuracy is below chance at initialization, about 61 percent at the window, and about 99 percent at maturity. Because the classifier orientation is fixed on the training split, below chance accuracy means the probe does not transfer in the orientation it learned, which is consistent with an orientation reversal rather than an absence of information, and its rise over training is what shows a transferable signal being learned.

Table 4. Held-out probe accuracy for predicting whether the S2 position holds a repeated name or a non-repeating control name, read from the S2 representation on Pythia 160M. Values are the mean over the two folds. The classifier orientation is fixed on the training split, so the below chance value at initialization reflects failure to transfer in that orientation.

	Initialization	Window (step 2000)	Maturity (final)
Layer 1	0.436	0.613	0.992
Layer 5	0.443	0.616	0.994

Position shuffle. This experiment measures how much of the initial in-distribution probe accuracy comes from architectural and positional structure. We shuffle the token positions before probing. Table 5 reports in-distribution probe accuracy on intact and position shuffled inputs, per layer, at initialization. Across these middle layers the intact probe reaches about 88 to 90 percent and shuffling reduces it by roughly 6 to 10 points, so a material part of the initial in-distribution accuracy is explained by fixed structure rather than by a learned signal.

Projection removal. This experiment is reported in Appendix H. It shows that removing the direction a duplication probe uses does not reduce the repeated name bias.

Table 5. In-distribution probe accuracy on intact and position shuffled inputs, per layer, on Pythia 160M at initialization. Shuffling reduces accuracy at every layer.

Layer	Intact	Position shuffled
1	88.7%	82.3%
2	89.0%	82.7%
3	90.0%	79.7%
6	87.7%	78.7%

D. Suppressor Trajectory and Split-Safe Confirmation

Table 6 traces the effect of mean-ablating the suppressor head across training, for Pythia 160M and Pythia 410M. In each model the head is the one whose mean ablation most reduces mature IOI performance, and the same head is mean-ablated at every earlier checkpoint without reselection. The effect is near zero during the window and grows during recovery, in step with the recovery of accuracy. The dominant head differs between the two models, which is expected because head positions are not shared across models. The five heads with the largest maturity effect in Pythia 160M are L8H9 with an effect of 2.66, followed by L3H0, L4H11, L6H2, and L1H8, with effects of 1.78, 1.54, 0.78, and 0.66.

Table 6. Change in logit difference from mean-ablating the mature suppressor head at each training step, with the accuracy at that step. The head is fixed at its maturity identity and mean-ablated at every earlier checkpoint.

Step	Pythia 160M (L8H9)		Pythia 410M (L12H12)	
	Mean-ablation effect	Accuracy	Mean-ablation effect	Accuracy
1000	−0.00	0.42	+0.00	0.45
2000	−0.01	0.32	−0.00	0.29
3000	−0.74	0.57	−0.17	0.82
5000	−2.39	0.88	−0.87	0.85
8000	−2.49	0.97	−1.62	0.98
13000	−1.77	0.97	−1.81	0.98
final	−2.66	1.00	−1.76	1.00

Split-safe confirmation. Because the head above is chosen using its mature effect, we confirm the ordering with two stricter analyses on the retrained model, and we keep them distinct because they use different evaluation sets and estimands.

The first freezes a mature-defined suppressor set by its functional signature on a selection split, then evaluates the effect of replacing that fixed set on 192 held-out items, giving 384 matched contrasts per checkpoint, with cluster-aware intervals. This set-level analysis uses matched replacement of the selected heads’ activations and is therefore a separate estimand from the single-head mean-ablation trajectory. Table 7 reports the fair three-way margin effect of this replacement across training. At steps 1800 and 2000 the effect is significantly positive, 0.091 at step 2000 with an interval from 0.062 to 0.121, so replacing the mature-defined set improves behavior and the set is not yet acting suppressively. The effect is indistinguishable from zero at steps 3200 and 5000 and is significantly negative by step 6000, at negative 0.64 with an interval from negative 0.80 to negative 0.47. These are read as an observed ordering rather than exact formation times, since the checkpoint at which an interval first excludes zero depends on effect size and variance.

The second uses a single mature-selected suppressor head, evaluated on a disjoint set of 800 examples with cluster-aware intervals. At step 2000 its ablation effect is 0.04, with an interval from negative 0.02 to 0.11, which includes zero, so this single-head also has no clearly detectable suppressive effect during the window. The frozen-set trajectory places robust suppressor function after the window, while the disjoint single-head analysis independently confirms that no clearly detectable suppressive effect is present at step 2000, so the developmental ordering is not an artifact of selecting a component on the same data on which its mature effect is reported.

Table 7. Frozen suppressor set on the retrained model, evaluated on 192 held-out items. Fair three-way margin effect of replacing the set, with cluster-aware 95 percent intervals. A negative value means that replacing the native suppressor set harms the fair three-way margin, so the native set is supporting the correct answer. Replacing the set significantly improves the margin during the window, has no clearly detectable effect at the intermediate checkpoints, and significantly harms the margin by step 6000.

Step	Margin effect	Cluster-aware interval
1800	+0.093	[+0.062, +0.125]
2000	+0.091	[+0.062, +0.121]
3200	−0.051	[−0.115, +0.014]
5000	−0.079	[−0.163, +0.006]
6000	−0.636	[−0.802, −0.468]

E. Behavior and Intervention Reversal Across Scale

Table 8 reports, for six Pythia scales, the lowest IOI accuracy during training and the matched S2 intervention effect at the window and at maturity, all measured under the ten template protocol used in the main analyses. The floor accuracy and the intervention effect in each row are measured in the same experiment on the same prompts. Every scale falls below chance during early training, and the intervention effect is positive during the window and negative at maturity at every scale. For the 160M primary window the effect survives dependence-aware inference. Resampling by template gives 0.68 to 1.25 at the dip and negative 4.54 to negative 3.72 at maturity, and resampling by name-pair gives 0.78 to 1.15 and negative 4.45 to negative 3.82. The intervention effects at the other scales are point estimates under the prompt-level protocol, and we do not claim that a cluster-aware interval excludes zero for the small 410M dip effect. The three smallest scales use windows of layers 3 to 5, 6 to 10, and 4 to 8 respectively. The larger models use middle layer windows scaled to depth, listed in the last column.

Table 8. Behavior and intervention reversal across six Pythia scales, all under the ten template protocol. Floor accuracy and intervention effect in each row are from the same experiment. The intervention effect at the window is positive and at maturity is negative at every scale.

Model	Floor accuracy	Intervention at window	Intervention at maturity	Patch window
Pythia 160M	31.7%	+0.95	−4.12	layers 3 to 5
Pythia 410M	29.3%	+0.11	−3.63	layers 6 to 10
Pythia 1B	36.3%	+0.43	−3.49	layers 4 to 8
Pythia 2.8B	29.7%	+0.33	−4.05	layers 8 to 14
Pythia 6.9B	32.3%	+0.84	−4.12	layers 8 to 14
Pythia 12B	42.0%	+0.43	−3.96	layers 9 to 16

F. Robustness Across Seeds, Data Order, and Initialization

Table 9 reports, for each of the nine PolyPythias 160M variants, the lowest accuracy during training and the matched S2 intervention effect at the window and at maturity, all under the ten template protocol. Three variants change the random seed, three change the order of the training data, and three change the initialization, each holding the other factors fixed. Every variant falls below chance, and the intervention effect reverses sign, positive at the window and negative at maturity, in all nine variants.

G. Stanford GPT-2 Intervention Reversal

The reversal also appears in a second model family. Table 10 reports the matched S2 intervention effect on Stanford GPT-2 Small across training. The effect is positive during the window and negative at maturity, the same pattern as in Pythia.

Table 9. Lowest IOI accuracy and matched S2 intervention effect for the nine PolyPythias 160M variants, under the ten template protocol. The intervention reverses sign in all nine.

Variant	Floor accuracy	Intervention at window	Intervention at maturity
Seed 1	30.7%	+0.55	−3.92
Seed 3	14.7%	+0.99	−2.89
Seed 5	34.3%	+1.11	−4.04
Data order 1	29.3%	+1.20	−4.06
Data order 2	28.0%	+1.35	−3.71
Data order 3	35.3%	+1.01	−3.24
Initialization 1	33.3%	+0.80	−4.56
Initialization 2	36.7%	+0.84	−4.75
Initialization 3	31.0%	+1.47	−3.51

Table 10. Matched S2 intervention effect on Stanford GPT-2 Small across training. The effect reverses from positive during the window to negative at maturity.

Step	1500	3000	10000	100000
Intervention effect	+1.03	+0.77	−0.23	−2.89

H. Removing the Probe Direction

This experiment tests whether the direction a probe uses to read the repeated name is the direction the model’s behavior depends on. We train a probe on the S2 representation at the window, then remove its direction at increasing strength and measure the effect on the logit difference. Table 11 also reports three control directions, a direction orthogonal to the probe direction, a direction from a probe trained on shuffled labels, and a random direction. Removing the probe direction does not reduce the preference for the repeated name at any strength. It slightly increases the preference, and the increase grows with strength, while the control directions have almost no effect. The direction that reads the repeated name is therefore not sufficient to explain the preference for it during the window.

Table 11. Effect of removing directions from the S2 representation at the window on Pythia 160M. Removing the probe direction does not reduce the preference for the repeated name at any strength. Lower accuracy means a stronger preference for the repeated name.

Direction removed	Logit difference	Accuracy
None (baseline)	−0.75	0.32
Probe direction, strength 0.5	−0.81	0.30
Probe direction, strength 1.0	−0.86	0.29
Probe direction, strength 2.0	−0.95	0.27
Probe direction, strength 4.0	−1.07	0.23
Orthogonal control, strength 1.0	−0.74	0.32
Shuffled label control, strength 1.0	−0.75	0.32
Random control, strength 1.0	−0.74	0.32

I. Reproducibility

The analyses use released Pythia checkpoints (Biderman et al., 2023), the nine PolyPythias 160M variants (van der Wal et al., 2025), and Stanford GPT-2 Small (Karamcheti et al., 2021), together with one Pythia 160M model trained from scratch with a new seed. Protocols differ by experiment, and we list them per experiment rather than stating one protocol for all.

- Matched S2 intervention. Ten single-token name templates in balanced ABBA and BABA order, 300 prompts. The S2 representation is replaced over a middle layer window, layers 3 to 5 for 160M, 6 to 10 for 410M, 4 to 8 for 1B, and

depth-scaled windows for the larger models, listed in Appendix E, with the representation from a control prompt whose second occurrence of the repeated name is a third single-token name.

- Position battery. The same 300 prompt set, perturbing one position at a time.
- Suppressor trajectory. The same 300 prompt set. Zero-ablation sets a head’s output to zero and mean-ablation replaces it with its average output over the set.
- Held-out probe. A probe trained on 600 examples from one set of names and templates and evaluated on a disjoint set, in two folds that swap the sets, predicting the repeated name position against a non-repeating control position.
- Position-shuffle probe. In-distribution probes on intact and position shuffled inputs.
- Projection removal. A probe trained on the S2 representation, whose direction is removed at several strengths.
- Split-safe suppressor set. A suppressor set fixed by functional signature on a selection split, evaluated on a disjoint held-out split.
- Locked input control. A locked benchmark of 800 prompts with a fixed prompt set and a stored prompt hash, comparing de-duplication, an alternate de-duplication, and a filler placebo, with cluster-aware intervals.
- Single-head suppressor analysis. A single mature-selected suppressor head evaluated on a disjoint set of 800 examples with cluster-aware intervals.
- Language modeling loss. A fixed sample from the Pile training stream on the monology Pile uncopyrighted train split (Gao et al., 2020), evaluated as described in Appendix A. The sampled texts are not recorded in the artifact.

Unless otherwise stated, intervals are 95 percent bootstrap intervals over prompts. Results explicitly labeled cluster-aware resample the recorded item, template, or name-pair clusters. The held-out probe reports point estimates over the evaluation examples. Code and analysis scripts, with a mapping from each figure and table to the script and result file that produced it, are available at a public repository linked from the camera-ready version.